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FINAL YEAR THESIS

A Fractional-Order Compartmental Model of HIV Transmission with Social Behaviour Interventions and Special Functions

Fractional Calculus | Special Functions | HIV Dynamics | Computational Simulation

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Declaration

I, NDACYAYISABA Lambert, declare that this final year thesis is my original work and has not been submitted elsewhere for any academic award. All sources used in this work have been acknowledged through proper citation and referencing.

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Approval

This thesis has been submitted for examination with my approval as the university supervisor.

Supervisor: Dr. MUHIRWA Jean Pierre

Signature: ______________________________ **Date:** __________________

Dedication

This work is dedicated to my family, mentors, lecturers, and all people who believe that mathematics can be used to understand and improve human life.

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Abstract

HIV/AIDS remains a major public health concern, particularly in sub-Saharan Africa, where transmission and control are shaped by both biomedical and social factors. Although antiretroviral therapy has improved survival and reduced transmission risk, HIV dynamics continue to be influenced by awareness, safer sexual behaviour, HIV testing, treatment-seeking behaviour, stigma, and adherence to medication. These factors suggest that HIV transmission cannot be fully understood through biological processes alone.

This thesis develops and analyses a fractional-order SITA model for HIV transmission with social behaviour interventions. The population is divided into four compartments: susceptible individuals $S(t)$, infected individuals $I(t)$, treated individuals $T(t)$, and individuals in the AIDS stage $A(t)$. The model is formulated using the Caputo fractional derivative of order q, where $0 < q \leq 1$. The fractional order introduces memory effects, allowing the model to reflect the fact that behavioural change, treatment adherence, and public health interventions may influence transmission over time rather than instantaneously.

Special functions, especially the Gamma function and the Mittag-Leffler function, are introduced as essential tools in the formulation and analysis of the fractional model. The Gamma function appears in the definition of the Caputo derivative, while the Mittag-Leffler function plays a role similar to the exponential function in fractional dynamical systems. The thesis presents positivity, boundedness, disease-free equilibrium, the basic reproduction number $\mathcal{R}_0$, and local stability using fractional-order stability conditions. Numerical simulations are structured using a fractional Adams–Bashforth–Moulton predictor-corrector method, and an interactive Python dashboard is designed to visualize memory effects, intervention scenarios, sensitivity analysis, and epidemic trajectories.

Core Contribution

This thesis extends a standard HIV compartmental model by replacing ordinary derivatives with Caputo fractional derivatives and by incorporating social behaviour interventions into transmission, treatment uptake, and progression mechanisms. The work connects fractional calculus, special functions, epidemiological modelling, and computational simulation.

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List of Abbreviations and Symbols

Symbol	Meaning
HIV	Human Immunodeficiency Virus
AIDS	Acquired Immunodeficiency Syndrome
ART	Antiretroviral Therapy
SITA	Susceptible–Infected–Treated–AIDS model
$S(t)$	Susceptible population at time t
$I(t)$	Infected untreated population at time t
$T(t)$	Treated population at time t
$A(t)$	AIDS-stage population at time t
$N(t)$	Total population at time t
q	Fractional order, $0 < q \leq 1$
${}^{C}D_t^q$	Caputo fractional derivative of order q
$\Gamma(z)$	Gamma function
$E_q(z)$	Mittag-Leffler function
$\mathcal{R}_0$	Basic reproduction number
$u_1(t)$	Awareness and education intervention
$u_2(t)$	Safer sexual behaviour / condom use intervention
$u_3(t)$	Testing and treatment-seeking intervention
$u_4(t)$	Treatment adherence intervention

Chapter 1

Introduction

1.1 Background and Motivation

HIV/AIDS remains a major public health problem whose transmission is shaped by biological, behavioural, and social factors. Treatment through ART has changed HIV from a highly fatal disease into a manageable chronic condition for many people, but HIV transmission continues to be affected by awareness, testing behaviour, disclosure, stigma, safer sexual practices, and adherence to treatment (WHO, 2024; UNAIDS, 2025).

Mathematical modelling provides a systematic way of understanding how HIV spreads and how interventions may influence long-term dynamics. Classical compartmental models are commonly formulated using ordinary differential equations. However, ordinary models assume that the rate of change depends only on the current state of the system. This assumption may be restrictive in HIV modelling because social behaviour and treatment response often carry memory: past awareness campaigns, previous adherence patterns, and earlier testing behaviour can continue to influence present transmission.

Fractional calculus provides a natural extension of ordinary calculus by allowing derivatives of non-integer order. A fractional-order model can capture memory and hereditary effects, making it suitable for diseases and interventions whose dynamics are influenced by past states (Podlubny, 1999; Kilbas et al., 2006; Diethelm, 2010). For this reason, this thesis formulates a fractional-order HIV transmission model using the Caputo fractional derivative.

1.2 Rwanda and Public Health Context

Rwanda has made important progress in HIV prevention, testing, treatment, and care. The Rwanda Population-based HIV Impact Assessment 2018–2019 reported adult HIV prevalence of approximately 3.0% among adults aged 15–64 years, annual adult incidence around 0.08%, and viral load suppression among HIV-positive adults around 76.0% (RBC, 2020). These findings show progress but also indicate that HIV transmission has not been fully eliminated.

Social behaviour remains relevant in the Rwandan HIV response. Stigma, testing behaviour, disclosure, and adherence can affect prevention and treatment pathways (RRP+, 2020). Therefore, a model that combines fractional dynamics with social behaviour interventions can provide useful academic insight into how memory and behavioural change may influence HIV transmission.

1.3 Statement of the Problem

Many existing HIV models use ordinary differential equations and treat transmission as a direct function of current infected populations. Such models may not fully capture memory effects arising from past behaviour, treatment adherence, community awareness, or stigma reduction. In addition, social behaviour is often represented as a fixed parameter rather than as intervention functions that modify transmission, treatment uptake, and progression.

This thesis addresses this limitation by developing a fractional-order SITA model that incorporates social behaviour interventions. The model uses the Caputo derivative to include memory effects and introduces intervention functions representing awareness, safer sexual behaviour, testing and treatment-seeking, and treatment adherence.

The central mathematical problem is therefore to connect three components in one coherent framework: a biologically meaningful HIV compartmental model, a fractional-order derivative that represents memory, and intervention functions that can be varied computationally. This connection is important because it allows the model to be analysed theoretically and then explored numerically through a dashboard.

1.4 Aim of the Study

The main aim of this thesis is to formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions and special functions.

1.5 Specific Objectives

1. To formulate a fractional-order SITA HIV transmission model using the Caputo fractional derivative.
2. To introduce social behaviour interventions representing awareness, safer sexual behaviour, testing and treatment-seeking, and treatment adherence.
3. To define the role of special functions, particularly the Gamma function and the Mittag-Leffler function, in the fractional model.
4. To establish positivity and boundedness of solutions.
5. To determine the disease-free equilibrium of the fractional model.
6. To derive the basic reproduction number $\mathcal{R}_0$ using the next-generation matrix method.
7. To analyse local stability using fractional-order stability criteria.
8. To perform numerical simulations for different fractional orders and intervention scenarios.
9. To develop an interactive Python dashboard for visualising model behaviour, memory effects, and intervention impact.

1.6 Research Questions

1. How can a SITA HIV transmission model be reformulated using Caputo fractional derivatives?
2. How do social behaviour interventions modify transmission, treatment uptake, and AIDS progression?

3. What role do the Gamma function and Mittag-Leffler function play in the fractional-order model?

4. What is the basic reproduction number $\mathcal{R}_0$ of the model?

5. Under what fractional-order stability conditions is the disease-free equilibrium locally stable?

6. How does varying the fractional order q affect HIV transmission dynamics?

7. How can a dashboard communicate the effect of memory and behavioural interventions in a clear and interactive way?

1.7 Significance of the Study

This study is significant in three ways. Mathematically, it introduces fractional calculus and special functions into HIV modelling and shows how the Caputo derivative, Gamma function, and Mittag-Leffler function enter the analysis of a biological system. Epidemiologically, it provides a structured framework for comparing behavioural and treatment-related interventions through the basic reproduction number and numerical trajectories. Computationally, it supports the development of an interactive dashboard that makes the fractional model testable, reproducible, and easier to communicate during presentation.

The work is also relevant to applied mathematics because it combines model formulation, qualitative analysis, numerical approximation, parameter-based simulation, and software implementation. These components show how mathematical theory can be translated into an exploratory tool for understanding disease dynamics.

1.8 Scope of the Study

The study focuses on a deterministic fractional-order SITA model with four compartments. It considers social behaviour interventions but does not include age structure, gender structure, spatial movement, stochasticity, or direct patient-level data. The dashboard is designed for academic simulation and interpretation, not clinical prediction or direct policy forecasting.

All numerical results should therefore be interpreted as scenario-based mathematical experiments. The purpose is to compare the qualitative effects of memory and intervention levels, not to estimate exact HIV prevalence or incidence in a real population.

1.9 Chapter Summary

This chapter introduced the study, motivated the use of fractional calculus for HIV modelling, and presented the problem statement, aim, objectives, research questions, significance, and scope. The next chapter reviews the mathematical and epidemiological literature supporting the thesis.

Chapter 2

Literature Review and Mathematical Preliminaries

2.1 Compartmental Models in Epidemiology

Compartmental epidemic modelling has its roots in the work of Kermack and McKendrick (1927), who introduced the classical SIR framework. Later studies extended such models to represent different infectious disease dynamics. Hethcote (2000) provides a comprehensive review of infectious disease models, including threshold quantities such as the basic reproduction number.

2.2 HIV/AIDS Mathematical Models

HIV differs from many acute infections because infected individuals do not naturally move into a permanently recovered class. For this reason, HIV models often include treatment and AIDS progression compartments (Anderson and May, 1991; Silva and Torres, 2017). In this thesis, the population is divided into susceptible, infected, treated, and AIDS classes.

2.3 Social Behaviour in HIV Transmission

HIV transmission is strongly influenced by behavioural and social factors. These include condom use, number of sexual partners, HIV testing, disclosure, stigma, treatment-seeking behaviour, and adherence to ART. Mathematical models can represent these effects through risk groups, behaviour-dependent transmission rates, or intervention functions (Massad et al., 2022; Supervie et al., 2013).

2.4 Fractional Calculus in Epidemiological Models

Fractional calculus extends ordinary differentiation and integration to non-integer orders. In epidemiological modelling, fractional derivatives are useful because they can incorporate memory and hereditary effects (Podlubny, 1999; Kilbas et al., 2006; Diethelm, 2010). Several fractional HIV/AIDS models have been proposed using Caputo and related operators (Günerhan et al., 2020; Unaegbu et al., 2021).

2.5 Gamma Function

The Gamma function is a fundamental special function in fractional calculus. It generalizes the factorial function and is defined by

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx, \qquad z > 0. \tag{2.1}$$

For positive integers n, it satisfies

$$\Gamma(n) = (n-1)!. \tag{2.2}$$

The Gamma function appears in the definition of fractional integrals and derivatives.

2.6 Riemann–Liouville Fractional Integral

For a function f and order $q > 0$, the Riemann–Liouville fractional integral is defined as

$$I_t^q f(t) = \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} f(s)\, ds. \tag{2.3}$$

This integral operator shows explicitly how past values of f contribute to the present value, weighted by a power-law memory kernel.

2.7 Caputo Fractional Derivative

For $0 < q < 1$, the Caputo fractional derivative of a differentiable function f is defined by

$$ {}^{C}D_t^q f(t) = \frac{1}{\Gamma(1-q)} \int_0^t \frac{f'(s)}{(t-s)^q}\, ds. \tag{2.4} $$

The Caputo derivative is used in this thesis because it allows classical initial conditions, such as $S(0) = S_0$, $I(0) = I_0$, $T(0) = T_0$, and $A(0) = A_0$.

Why the Caputo Derivative is Selected

The Caputo derivative is selected because it is suitable for biological models with standard initial conditions. It also carries a memory kernel, meaning that the present dynamics depend on past states. This is important for HIV modelling because awareness, adherence, stigma reduction, and testing behaviour may influence transmission over time rather than only at the present moment.

2.8 Mittag-Leffler Function

The Mittag-Leffler function is a special function that generalizes the exponential function in fractional calculus. The one-parameter Mittag-Leffler function is defined by

$$ E_q(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(qk+1)}, \qquad q > 0. \tag{2.5} $$

The two-parameter form is

$$ E_{q,r}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(qk+r)}, \qquad q, r > 0. \tag{2.6} $$

When $q = 1$, the Mittag-Leffler function reduces to the exponential function:

$$ E_1(z) = e^z. \tag{2.7} $$

This makes the Mittag-Leffler function central in fractional differential equations, where it often replaces the exponential function in solutions and stability analysis (Kilbas et al., 2006; Diethelm, 2010).

2.9 Fractional Stability Criterion

For a linear fractional system

$$ {}^{C}D_t^q x(t) = Ax(t), \qquad 0 < q \leq 1, \tag{2.8} $$

the equilibrium is locally asymptotically stable if every eigenvalue λ_i of A satisfies

$$ |\arg(\lambda_i)| > \frac{q\pi}{2}. \tag{2.9} $$

This criterion is more general than the ordinary condition $\mathrm{Re}(\lambda_i) < 0$ and becomes the classical condition when $q = 1$ (Matignon, 1996).

2.10 Summary of Literature Gap

The literature shows that compartmental HIV models are useful, fractional derivatives can represent memory effects, and special functions provide the mathematical foundation for fractional models. However, there remains room for a model that combines fractional-order HIV dynamics with social behaviour interventions and a computational dashboard. This thesis addresses that gap.

2.11 Chapter Summary

This chapter reviewed compartmental epidemic models, HIV/AIDS mathematical models, social behaviour in HIV transmission, and the fractional calculus tools needed for the study. The next chapter formulates the fractional-order SITA model with social behaviour interventions.

Chapter 3

Fractional-Order Model Formulation

3.1 Population Compartments

The total population at time t is divided into four compartments:

Compartment	Description
$S(t)$	Susceptible individuals who are not infected but are at risk of HIV infection.
$I(t)$	Infected individuals not yet on treatment.
$T(t)$	HIV-positive individuals receiving ART.
$A(t)$	Individuals in the AIDS stage.

The total population is

$$N(t) = S(t) + I(t) + T(t) + A(t). \tag{3.1}$$

3.2 Model Assumptions

The model is formulated under the following assumptions:

1. The population is divided into four mutually exclusive compartments: susceptible, infected, treated, and AIDS.
2. Individuals enter the susceptible class through recruitment at rate Λ.
3. Susceptible individuals acquire HIV through effective contact with infected or treated individuals.

4. Treated individuals have reduced infectiousness compared with untreated infected individuals, represented by η, where $0 < \eta < 1$.

5. Infected individuals may begin treatment and move into the treated class.

6. Infected individuals may progress to AIDS at rate δ.

7. Treated individuals may progress to AIDS at rate ρ, but adherence intervention reduces this progression.

8. Natural mortality occurs in all compartments at rate μ.

9. AIDS-induced mortality occurs in the AIDS compartment at rate d.

10. Social behaviour interventions are bounded controls satisfying $0 \leq u_i(t) \leq 1$, $i = 1, 2, 3, 4$.

11. The fractional order satisfies $0 < q \leq 1$, where $q = 1$ gives the classical ordinary model and $0 < q < 1$ introduces memory.

12. During the analytical derivation of $\mathcal{R}_0$, the intervention levels are treated as constants so that the disease-free equilibrium and next-generation matrix can be defined clearly.

13. The model is intended for qualitative and comparative analysis; parameter values used in simulations represent baseline academic scenarios rather than direct clinical calibration.

These assumptions simplify the complex social and biological processes involved in HIV transmission. They are useful for constructing a mathematically tractable model, but they also define the limitations of the conclusions drawn from the simulations.

3.3 Social Behaviour Intervention Functions

The model includes four social behaviour interventions:

Intervention	Meaning	Range
$u_1(t)$	Awareness and education intervention	$0 \leq u_1(t) \leq 1$
$u_2(t)$	Condom use / safer sexual behaviour intervention	$0 \leq u_2(t) \leq 1$
$u_3(t)$	Testing and treatment-seeking intervention	$0 \leq u_3(t) \leq 1$
$u_4(t)$	Treatment adherence intervention	$0 \leq u_4(t) \leq 1$

3.4 Effective Transmission and Transition Terms

The effective transmission rate is defined as

$$\beta_{\text{eff}}(t) = \beta_0(1 - u_1(t))(1 - u_2(t)). \tag{3.2}$$

The force of infection is

$$\lambda(t) = \frac{\beta_{\text{eff}}(t)(I + \eta T)}{N}, \tag{3.3}$$

where $0 < \eta < 1$ represents the reduced infectiousness of treated individuals.

The effective treatment uptake and treated-to-AIDS progression rates are

$$\tau_{\text{eff}}(t) = \tau(1 + u_3(t)), \tag{3.4}$$

and

$$\rho_{\text{eff}}(t) = \rho(1 - u_4(t)). \tag{3.5}$$

3.5 Fractional-Order SITA Model

Using the Caputo fractional derivative of order q, $0 < q \leq 1$, the proposed model is

$$^{C}D_t^q S(t) = \Lambda - \lambda(t)S - \mu S, \tag{3.6}$$

$$^{C}D_t^q I(t) = \lambda(t)S - [\tau(1 + u_3(t)) + \delta + \mu]I, \tag{3.7}$$

$$^{C}D_t^q T(t) = \tau(1 + u_3(t))I - [\rho(1 - u_4(t)) + \mu]T, \tag{3.8}$$

$$^{C}D_t^q A(t) = \delta I + \rho(1 - u_4(t))T - (\mu + d)A. \tag{3.9}$$

The initial conditions are

$$S(0) = S_0, \qquad I(0) = I_0, \qquad T(0) = T_0, \qquad A(0) = A_0. \tag{3.10}$$

Connection with the Ordinary Model

When $q = 1$, the Caputo derivative becomes the ordinary first derivative and the fractional-order model reduces to the classical SITA ordinary differential equation model. When $0 < q < 1$, the model includes memory effects.

3.6 Model Flow Diagram

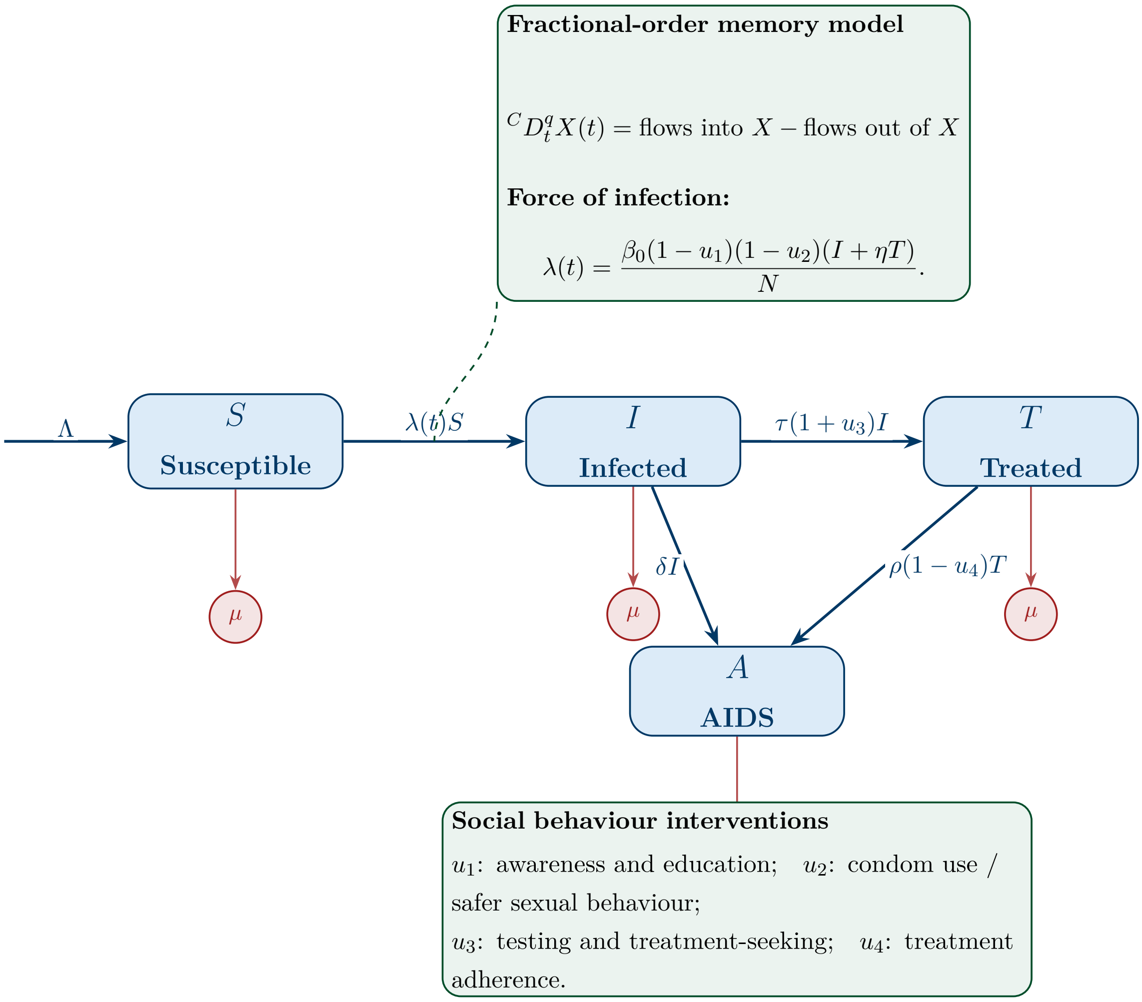


Figure 3.1: Flow diagram of the fractional-order SITA HIV transmission model with social behaviour interventions. The compartmental structure is similar to a classical HIV model, but the dynamics are governed by Caputo fractional derivatives, allowing the model to include memory effects.

3.7 Parameter Description

Parameter	Description	Range
Λ	Recruitment rate into susceptible class	> 0
μ	Natural mortality rate	> 0
β_0	Baseline HIV transmission rate	> 0
η	Relative infectiousness of treated individuals	$0 < \eta < 1$
τ	Baseline treatment initiation rate	> 0
δ	Progression rate from infected class to AIDS	> 0
ρ	Progression rate from treated class to AIDS	> 0
d	AIDS-induced mortality rate	> 0
q	Fractional order representing memory effect	$0 < q \leq 1$
$u_1(t)$	Awareness and education intervention	$0 \leq u_1 \leq 1$
$u_2(t)$	Safer sexual behaviour intervention	$0 \leq u_2 \leq 1$
$u_3(t)$	Testing and treatment-seeking intervention	$0 \leq u_3 \leq 1$
$u_4(t)$	Treatment adherence intervention	$0 \leq u_4 \leq 1$

Parameter Use and Interpretation

The parameters in this thesis are used for mathematical simulation and scenario comparison. Some parameters, such as mortality and HIV prevalence context, are motivated by public-health literature, while other transition and intervention parameters are selected as baseline simulation values. The aim is not to claim exact calibration to Rwanda-specific individual data, but to study how the fractional order and intervention controls affect the qualitative behaviour of the model.

Parameter	Role in the model	Baseline source status
Λ	Recruitment into the susceptible population	Simulation/normalisation choice
μ	Natural mortality rate	Estimated or assumed
β_0	Baseline transmission intensity	Literature-guided/assumed
η	Relative infectiousness under treatment	Literature-guided/assumed
τ	Treatment initiation rate	Literature-guided/assumed
δ	Progression from infection to AIDS	Literature-guided/assumed
ρ	Progression from treatment to AIDS	Literature-guided/assumed
d	AIDS-induced mortality	Literature-guided/assumed
q	Fractional memory order	Simulation parameter
u_1, u_2, u_3, u_4	Social behaviour intervention levels	User-defined controls

3.8 Chapter Summary

This chapter formulated the fractional-order SITA model, defined the intervention functions, stated the assumptions, and presented the model diagram and parameters. The next chapter analyses the mathematical properties of the model.

Chapter 4

Theoretical Analysis of the Fractional Model

4.1 Equivalent Integral Form

Using the Riemann–Liouville fractional integral, the Caputo system can be written in equivalent integral form. For example,

$$S(t) = S_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} \left[\Lambda - \lambda(s)S(s) - \mu S(s)\right] ds. \tag{4.1}$$

Similar integral forms hold for $I(t)$, $T(t)$, and $A(t)$. This representation is important for proving existence, uniqueness, positivity, and for deriving numerical schemes.

4.2 Existence and Uniqueness of Solutions

Let

$$X(t) = (S(t), I(t), T(t), A(t))^T$$

and write the system in compact form as

$${}^{C}D_t^q X(t) = F(t, X(t)).$$

On the biologically feasible region, the denominator $N(t)$ is positive whenever the total population is positive. Hence the right-hand side consists of continuous functions of the state variables and is locally Lipschitz on bounded subsets of the positive region.

If $F(t, X)$ is continuous in t and locally Lipschitz continuous in X on the feasible region, then the fractional-order HIV model has a unique solution on a suitable time interval.

Proof Outline. Using the definition of the Caputo derivative, the fractional system is transformed into a Volterra-type integral equation:

$$X(t) = X(0) + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} F(s, X(s)) ds.$$

Under the continuity and local Lipschitz conditions stated above, the associated integral operator is a contraction on a suitable function space over a sufficiently small interval. Banach's fixed point theorem then guarantees the existence and uniqueness of a local solution. Since the population is shown below to remain bounded in the feasible region, the local solution can be extended while it remains in that region (Kilbas et al., 2006; Diethelm, 2010). □

4.3 Positivity of Solutions

If $S_0, I_0, T_0, A_0 \geq 0$, then

$$S(t), I(t), T(t), A(t) \geq 0, \qquad t \geq 0.$$

Proof Outline. At the boundary of the nonnegative region, the right-hand side of each compartment equation is nonnegative in the corresponding boundary component. Specifically, if $S = 0$, then ${}^C D_t^q S = \Lambda \geq 0$; if $I = 0$, then ${}^C D_t^q I = \lambda(t) S \geq 0$; if $T = 0$, then ${}^C D_t^q T = \tau(1 + u_3) I \geq 0$; and if $A = 0$, then ${}^C D_t^q A = \delta I + \rho(1 - u_4) T \geq 0$. By the positivity principle for Caputo fractional systems, solutions starting in the nonnegative orthant remain nonnegative. □

4.4 Boundedness and Feasible Region

Let

$$N(t) = S(t) + I(t) + T(t) + A(t).$$

Adding the model equations gives

$$^{C}D_t^q N(t) = \Lambda - \mu N(t) - dA(t) \leq \Lambda - \mu N(t). \tag{4.2}$$

The associated comparison fractional equation is

$$^{C}D_t^q Y(t) = \Lambda - \mu Y(t).$$

Its solution can be expressed using the Mittag-Leffler function:

$$Y(t) = \frac{\Lambda}{\mu} + \left(Y(0) - \frac{\Lambda}{\mu}\right) E_q(-\mu t^q).$$

Since $E_q(-\mu t^q)$ decays to zero for $\mu > 0$, it follows that

$$\limsup_{t\to\infty} N(t) \leq \frac{\Lambda}{\mu}.$$

Thus, the biologically feasible region is

$$\Omega = \left\{(S, I, T, A) \in \mathbb{R}_+^4 : 0 \leq N(t) \leq \frac{\Lambda}{\mu}\right\}. \tag{4.3}$$

This boundedness result also supports the biological interpretation of the model, because the total population remains finite and the compartments stay inside a positively invariant region.

4.5 Disease-Free Equilibrium

At the disease-free equilibrium, $I = T = A = 0$. Hence,

$$0 = \Lambda - \mu S,$$

and therefore

$$E_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0\right). \tag{4.4}$$

4.6 Basic Reproduction Number

For equilibrium analysis, the intervention functions are treated as constant intervention levels. Define

$$\beta_{\text{eff}} = \beta_0(1-u_1)(1-u_2), \qquad \tau_{\text{eff}} = \tau(1+u_3), \qquad \rho_{\text{eff}} = \rho(1-u_4).$$

Using the next-generation matrix method (Van den Driessche and Watmough, 2002), the infected compartments are I and T. The next-generation matrices are

$$F = \begin{pmatrix} \beta_{\text{eff}} & \eta\beta_{\text{eff}} \\ 0 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} \tau_{\text{eff}} + \delta + \mu & 0 \\ -\tau_{\text{eff}} & \rho_{\text{eff}} + \mu \end{pmatrix}.$$

The basic reproduction number is

$$\mathcal{R}_0 = \frac{\beta_{\text{eff}}}{\tau_{\text{eff}} + \delta + \mu}\left(1 + \frac{\eta\tau_{\text{eff}}}{\rho_{\text{eff}} + \mu}\right). \tag{4.5}$$

The first factor represents the expected contribution of untreated infected individuals before leaving the infected class. The second factor accounts for the additional transmission contribution from treated individuals, reduced by the relative infectiousness parameter η and by the treated-class exit rate. Substituting the intervention terms gives

$$\mathcal{R}_0 = \frac{\beta_0(1-u_1)(1-u_2)}{\tau(1+u_3) + \delta + \mu}\left(1 + \frac{\eta\tau(1+u_3)}{\rho(1-u_4) + \mu}\right). \tag{4.6}$$

4.7 Fractional-Order Local Stability

The disease-free equilibrium is locally asymptotically stable for the fractional model if all eigenvalues λ_i of the linearized infected subsystem at E_0 satisfy

$$|\arg(\lambda_i)| > \frac{q\pi}{2}. \tag{4.7}$$

When $q = 1$, this reduces to the classical ordinary differential equation stability condition.

For this model, $\mathcal{R}_0 < 1$ means that the expected number of secondary infections generated by one infected individual is below the epidemic threshold. The fractional stability condition then verifies that the memory-dependent linearized system still moves away from infection persistence and toward the disease-free equilibrium.

If $\mathcal{R}_0 < 1$ and the eigenvalues of the infected subsystem satisfy

$$|\arg(\lambda_i)| > \frac{q\pi}{2},$$

then the disease-free equilibrium E_0 is locally asymptotically stable in the fractional-order sense.

4.8 Sensitivity Analysis

The normalized sensitivity index of $\mathcal{R}_0$ with respect to a parameter p is

$$\Upsilon_p^{\mathcal{R}_0} = \frac{\partial \mathcal{R}_0}{\partial p} \cdot \frac{p}{\mathcal{R}_0}. \tag{4.8}$$

This is used to determine which biological parameters and social behaviour interventions have the strongest effect on HIV transmission.

4.9 Chapter Summary

This chapter presented the equivalent integral form, existence and uniqueness, positivity, boundedness, disease-free equilibrium, reproduction number, fractional stability criterion, and sensitivity analysis. The next chapter explains the research methodology, numerical scheme, and dashboard development process.

Chapter 5

Research Methodology, Numerical Method and Dashboard Development

5.1 Research Design

This thesis uses a mathematical and computational research design. The mathematical component formulates and analyses the fractional-order SITA model as presented in Chapters 3 and 4. The computational component implements a fractional numerical solver and an interactive web-based dashboard to explore intervention scenarios, memory effects, sensitivity analysis, and epidemic trajectories.

The research proceeds in two parallel streams. The analytical stream establishes the theoretical properties of the model: positivity, boundedness, disease-free equilibrium, basic reproduction number, and fractional stability. The computational stream translates these results into a working simulation environment that allows parameter exploration and visual verification of the analytical findings.

5.2 Model Development Procedure

The model development follows these steps:

1. Define the four SITA compartments: susceptible $S(t)$, infected $I(t)$, treated $T(t)$, and AIDS-stage $A(t)$.

2. Introduce four social behaviour intervention functions $u_1, u_2, u_3, u_4 \in [0, 1]$.

3. Replace ordinary derivatives with the Caputo fractional derivative of order q, $0 < q \leq 1$.

4. Derive the effective transmission rate β_{eff}, force of infection $\lambda(t)$, effective treatment uptake τ_{eff}, and effective AIDS progression rate ρ_{eff}.
5. Analyse positivity, boundedness, disease-free equilibrium, basic reproduction number $\mathcal{R}_0$, and fractional-order local stability.
6. Implement the fractional Adams–Bashforth–Moulton predictor-corrector solver in Python.
7. Build the interactive Flask dashboard with API endpoints, Plotly visualisations, and export tools.

5.3 Fractional Numerical Scheme

The numerical solution of the fractional-order SITA system is obtained using the fractional Adams–Bashforth–Moulton (ABM) predictor-corrector method (Diethelm et al., 2002; Diethelm, 2010). This method is well suited for Caputo fractional differential equations because it approximates the convolution integral that defines the Caputo derivative, thereby capturing the memory effect that all previous time steps contribute to the current state.

For a general Caputo fractional system

$$ {}^{C}D_t^q y(t) = f(t, y(t)), \qquad y(0) = y_0, \qquad 0 < q \leq 1, $$

the equivalent Volterra integral form is

$$ y(t) = y_0 + \frac{1}{\Gamma(q)} \int_0^t (t-s)^{q-1} f(s, y(s))\, ds. $$

Let $t_n = nh$ for $n = 0, 1, \dots, M$, where $h > 0$ is the fixed step size.

5.3.1 Predictor Step (Adams–Bashforth)

The predictor approximation at step $n+1$ is

$$ y_{n+1}^P = y_0 + \frac{h^q}{\Gamma(q+1)} \sum_{j=0}^{n} b_{j,n+1}\, f(t_j, y_j), \tag{5.1} $$

where the predictor weights are

$$ b_{j,n+1} = (n+1-j)^q - (n-j)^q, \qquad j = 0, 1, \dots, n. \tag{5.2} $$

5.3.2 Corrector Step (Adams–Moulton)

The corrector step uses the predicted value y_{n+1}^P to improve the approximation:

$$y_{n+1} = y_0 + \frac{h^q}{\Gamma(q+2)} \left[f\left(t_{n+1}, y_{n+1}^P\right) + \sum_{j=0}^{n} a_{j,n+1} f(t_j, y_j) \right], \tag{5.3}$$

where the corrector weights $a_{j,n+1}$ are derived from the fractional memory kernel:

$$a_{j,n+1} = \begin{cases} (n-1)^{q+1} - (n-1-q)\, n^q, & j = 0, \\ (n-j+2)^{q+1} + (n-j)^{q+1} - 2(n-j+1)^{q+1}, & 1 \le j \le n, \\ 1, & j = n+1. \end{cases} \tag{5.4}$$

Numerical Accuracy Note

The ABM method provides improved accuracy over a simple fractional Euler scheme. The order of convergence depends on the step size h, the fractional order q, and the smoothness of the solution. For the simulations in this thesis, a default step size of $h = 0.2$ years is used, which balances computational efficiency with numerical accuracy. Smaller step sizes improve accuracy but increase computation time due to the $O(M^2)$ memory accumulation in the corrector weights. Non-negativity constraints are applied at each step to ensure biological consistency.

5.4 Simulation Scenarios

Eight simulation scenarios are constructed to evaluate how social behaviour interventions and fractional memory modify the dynamics of the SITA HIV model. The scenarios are not intended to represent exact field measurements; rather, they provide a controlled numerical experiment for comparing the effect of each intervention pathway. Across the scenarios, the biological parameters $\Lambda, \mu, \beta_0, \tau, \delta, \rho, \eta$, and d are kept at their baseline values. Only the intervention controls u_1, u_2, u_3, u_4 and, where necessary, the fractional order q are varied.

The intervention controls are interpreted as follows: u_1 represents awareness and education, u_2 represents safer sexual behaviour, u_3 represents testing and treatment-seeking, and u_4 represents adherence support. The controls enter the model through the effective transmission, treatment, and AIDS-progression rates:

$$\beta_{\text{eff}} = \beta_0(1-u_1)(1-u_2), \qquad \tau_{\text{eff}} = \tau(1+u_3), \qquad \rho_{\text{eff}} = \rho(1-u_4).$$

Thus, larger values of u_1 and u_2 reduce new infections, larger values of u_3 increase movement from the infected class to the treated class, and larger values of u_4 reduce progression from treatment to AIDS.

Table 5.1: Simulation scenarios used for intervention and memory comparison.

No.	Scenario	u_1	u_2	u_3	u_4	q	Main purpose
1	No intervention	0.0	0.0	0.0	0.0	0.95	Reference trajectory for comparison with intervention cases.
2	Awareness only	0.7	0.0	0.0	0.0	0.95	Measures the isolated effect of education and awareness on transmission.
3	Safer behaviour only	0.0	0.7	0.0	0.0	0.95	Measures the isolated effect of condom use and safer sexual practices.
4	Testing only	0.0	0.0	0.7	0.0	0.95	Examines faster movement from infection to treatment.
5	Adherence only	0.0	0.0	0.0	0.8	0.95	Examines reduced progression from treatment to AIDS.
6	Combined intervention	0.7	0.7	0.6	0.8	0.95	Represents an integrated prevention, testing, treatment, and adherence strategy.
7	Ordinary model	0.5	0.5	0.5	0.5	1.00	Removes fractional memory to compare with the classical model.
8	High memory effect	0.4	0.4	0.4	0.4	0.75	Tests the effect of stronger memory on long-term dynamics.

Reference and Single-Intervention Scenarios

Scenario 1 gives the no-intervention reference case. Scenarios 2–5 then activate one control at a time while keeping the remaining controls equal to zero. This structure makes it possible to identify the individual contribution of awareness, safer behaviour, testing, and adherence before they are combined. Awareness and safer behaviour act directly on the effective transmission rate β_{eff}, testing modifies the effective treatment-seeking rate τ_{eff}, and adherence modifies the effective AIDS-progression rate ρ_{eff}.

Combined Intervention Scenario

Scenario 6 applies all four controls simultaneously. This scenario is important because HIV control in practice requires more than one action. Prevention reduces new infections, testing identifies infected individuals, treatment reduces untreated infection, and adherence reduces progression to AIDS. The combined scenario therefore represents a comprehensive control strategy and is expected to produce a stronger reduction in $\mathcal{R}_0$ than the single-intervention cases.

Memory-Comparison Scenarios

Scenarios 7 and 8 are included to compare classical and fractional dynamics. Scenario 7 sets $q = 1$, which removes the fractional memory effect and reduces the system to an ordinary differential equation model. Scenario 8 uses $q = 0.75$, which introduces stronger memory. Comparing these cases helps determine whether past epidemic states and delayed behavioural effects change the predicted trajectories.

This scenario design supports three levels of analysis. First, the no-intervention case establishes a reference curve. Second, the single-intervention cases identify which control pathways are most influential. Third, the combined and memory scenarios examine whether integrated intervention and fractional memory produce substantially different long-term outcomes.

5.5 Fractional-Order Comparison

To isolate the effect of the fractional order on epidemic dynamics, the model is simulated at four values of q under identical initial conditions and parameter values:

$$q = 1.00, \quad q = 0.95, \quad q = 0.85, \quad q = 0.75.$$

When $q = 1$, the model reduces to the classical ordinary differential equation system and the Mittag–Leffler function $E_q(-\mu t^q)$ reduces to the exponential function $e^{-\mu t}$. This case represents a memoryless model in which the present rate of change depends only on the current state variables. When $0 < q < 1$, the Caputo derivative introduces a memory kernel, meaning that past states contribute to the present dynamics.

As q decreases below one, the memory effect becomes stronger. In epidemiological terms, this means that past infection levels, treatment history, adherence behaviour, and accumulated behavioural change may continue to influence the present trajectory. Therefore, the fractional-order comparison is useful for determining whether the classical model underestimates or overestimates the persistence of infection, treatment burden, and AIDS progression.

The memory comparison is visualised in the dashboard Memory Effect tab. The dashboard plots $I(t)$, $T(t)$, $A(t)$, and the I–T phase trajectory for all four values of q simultaneously. It also includes a Mittag–Leffler versus exponential decay chart to show how the fractional memory kernel decays more slowly than the ordinary exponential kernel. These visualisations provide the numerical basis for the fractional-order results presented in Chapter 6.

5.6 Python Dashboard Architecture

The dashboard is developed using Python, Flask, HTML, CSS, JavaScript, and Plotly. It contains the following layers:

1. **Model layer**: implements the fractional SITA equations.
2. **Numerical solver layer**: implements the fractional predictor-corrector method.
3. **Analysis layer**: computes $\mathcal{R}_0$, sensitivity indices, peak infection, and final states.
4. **API layer**: exposes simulation, scenario comparison, sensitivity analysis, reproduction-number, and export endpoints.
5. **User interface layer**: provides sliders, scenario selectors, graphs, interpretation cards, and export tools using HTML, CSS, JavaScript, and Plotly.

5.7 Dashboard Features

- Slider for fractional order q.
- Sliders for biological parameters and intervention functions.
- Real-time computation of $\mathcal{R}_0$.
- Time-series plots for $S(t)$, $I(t)$, $T(t)$, $A(t)$, and $N(t)$.
- Scenario comparison plots.
- Sensitivity analysis charts.
- Memory-effect comparison for different values of q.

- Export of simulation results as CSV.
- Downloadable figures for thesis reporting.

5.8 Suggested Project Folder Structure

```
fractional_hiv_thesis_dashboard/
|
|-- app.py
|-- config.py
|-- requirements.txt
|-- README.md
|
|-- model/
|    |-- fractional_sita_model.py
|    |-- fractional_solver.py
|    |-- reproduction_number.py
|    |-- scenarios.py
|    |-- sensitivity.py
|    `-- validation.py
|
|-- routes/
|    |-- simulation_routes.py
|    |-- export_routes.py
|    `-- health_routes.py
|
|-- templates/
|    |-- base.html
|    |-- index.html
|    |-- dashboard.html
|    |-- documentation.html
|    `-- about.html
|
|-- static/
|    |-- css/
|    |-- js/
|    |-- assets/
|    `-- vendor/
|
|-- data/
```

```
|    |-- baseline_parameters.json
|    |-- scenario_presets.json
|    `-- sample_outputs.csv
|
`-- outputs/
     |-- csv/
     |-- figures/
     `-- reports/
```

5.9 Chapter Summary

This chapter described the research design, numerical scheme, simulation scenarios, fractional-order comparison, and dashboard architecture. The next chapter presents the expected results and discussion structure.

Chapter 6

Results and Discussion

6.1 Baseline Fractional Simulation

This section presents the baseline numerical simulation of the fractional-order SITA model using the parameter values and initial conditions described in the methodology chapter. The baseline simulation was performed for 50 years with step size $h = 0.2$ and fractional order $q = 0.95$. The initial population was

$$S(0) = 10000, \qquad I(0) = 150, \qquad T(0) = 80, \qquad A(0) = 20.$$

The social behaviour intervention levels used in the baseline dashboard configuration were

$$u_1 = 0.40, \qquad u_2 = 0.50, \qquad u_3 = 0.60, \qquad u_4 = 0.70.$$

These values represent moderate-to-strong awareness, safer behaviour, testing and treatment-seeking, and adherence support.

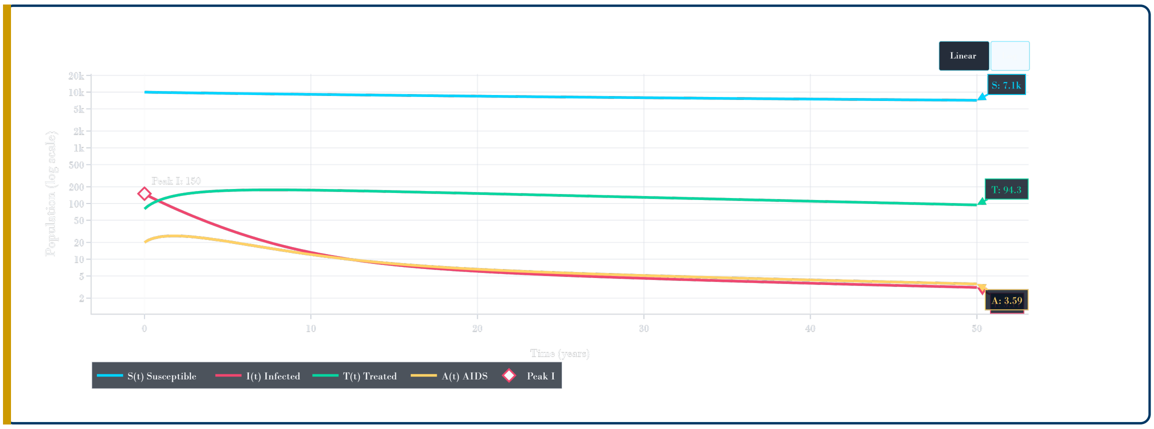
Under this configuration, the effective rates were

$$\beta_{\text{eff}} = 0.0900, \qquad \tau_{\text{eff}} = 0.3200, \qquad \rho_{\text{eff}} = 0.0090.$$

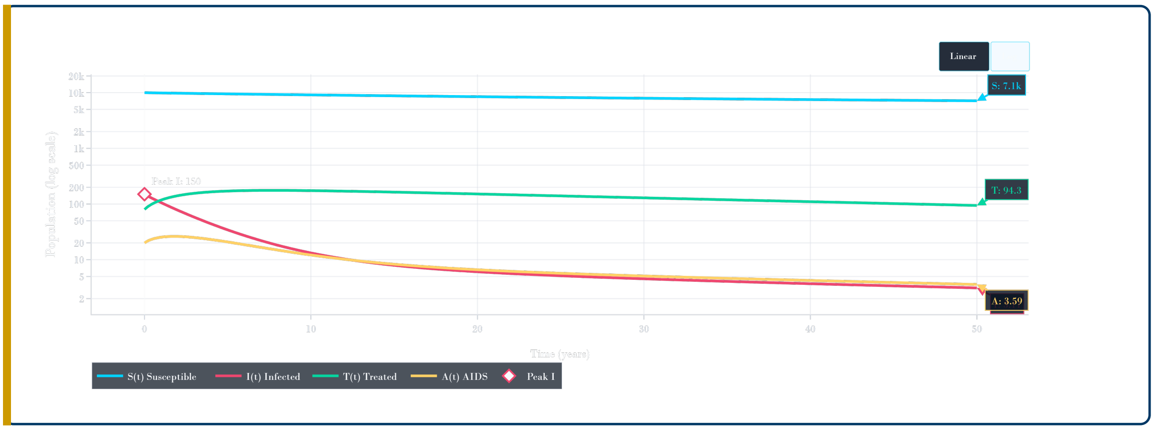
The computed basic reproduction number was

$$\mathcal{R}_0 = 0.430,$$

which is less than one. This indicates that, under the selected intervention-adjusted parameter values, the disease-free equilibrium is expected to be locally stable and the infection is expected to decline over time.



(a) Baseline SITA time-series graph in the dashboard linear view.



(b) Baseline SITA time-series graph in logarithmic view.

Figure 6.1: Baseline fractional-order SITA simulation for $q = 0.95$ showing $S(t)$, $I(t)$, $T(t)$, and $A(t)$.

Figure 6.1 shows the baseline behaviour of the four model compartments. The susceptible population $S(t)$ remains the largest compartment throughout the simulation and decreases gradually from the initial value of 10000 to approximately 7146.82 at the final time. The infected population $I(t)$ starts at its highest value of 150 and then decreases continuously, reaching approximately 3.11 by year 50. This decline is consistent with the computed value $\mathcal{R}_0 = 0.430 < 1$, which indicates controlled transmission under the selected intervention levels.

The treated class $T(t)$ initially increases because infected individuals are moved into treatment through the testing and treatment-seeking intervention. After reaching a higher level, it gradually decreases and ends at approximately 94.27. The AIDS-stage class $A(t)$ remains low compared with the susceptible and treated compartments and ends at approximately 3.59. The logarithmic view is especially useful because it makes the smaller infected, treated, and AIDS-stage curves visible on the same graph as the much larger susceptible population.

Table 6.1: Baseline simulation summary for the fractional SITA model.

Quantity	Value	Interpretation
$\mathcal{R}_0$	0.430	Controlled epidemic status
Peak infected	150.00	Occurs at $t = 0.0$ years
Final susceptible, $S(50)$	7146.82	Susceptible population at final time
Final infected, $I(50)$	3.11	Infected population at final time
Final treated, $T(50)$	94.27	Treated population at final time
Final AIDS, $A(50)$	3.59	AIDS-stage population at final time

The baseline results show a strong decline in the infected class. The infected population starts at 150 individuals and decreases to approximately 3.11 individuals by the end of the simulation period. The treated class remains higher than the infected and AIDS classes at the final time, which is consistent with the effect of testing and treatment-seeking intervention. The AIDS-stage population also remains low, which reflects the effect of adherence support in reducing progression from treatment to AIDS.

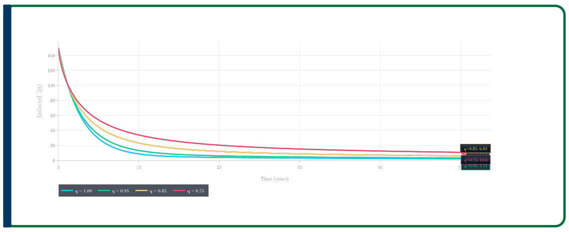
6.2 Effect of Fractional Order

The fractional order q controls the strength of memory in the Caputo fractional derivative. When $q = 1$, the model reduces to the classical ordinary differential equation model. When $0 < q < 1$, the model contains memory effects, meaning that previous states of the system continue to influence the present dynamics through the fractional kernel.

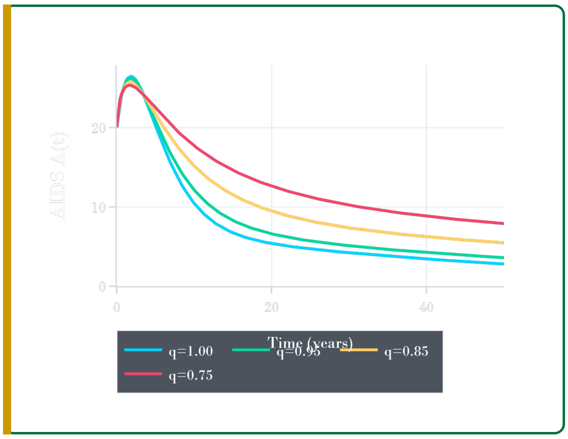
To examine the role of memory, the model was simulated using four values of the fractional order:

$$q = 1.00, \qquad q = 0.95, \qquad q = 0.85, \qquad q = 0.75.$$

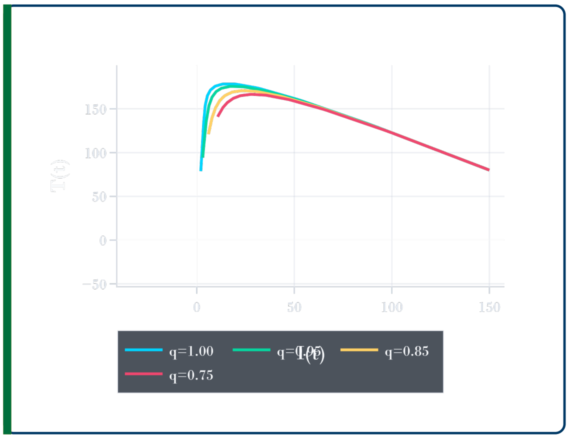
All other parameter values were kept fixed so that the differences in the trajectories could be attributed to the change in q.



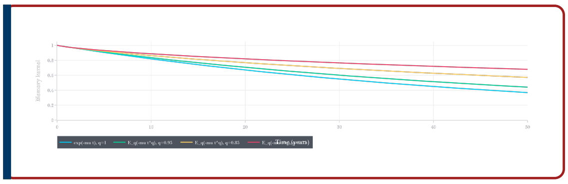
(a) Infected population $I(t)$.



(b) AIDS-stage population $A(t)$.



(c) Phase trajectory in the I–T plane.



(d) Mittag–Leffler memory kernel.

Figure 6.2: Two-by-two memory-effect dashboard visualisation for $q = 1.00$, $q = 0.95$, $q = 0.85$, and $q = 0.75$.

Figure 6.2 compares the model response for $q = 1.00$, $q = 0.95$, $q = 0.85$, and $q = 0.75$ in a compact two-by-two layout. In Figure 6.2a, all infected curves begin from the same initial value, but their decline rates are different. The ordinary model with $q = 1.00$ decreases fastest, while the lower fractional order $q = 0.75$ decreases slowest and remains highest at the final time. This confirms that stronger fractional memory delays the movement of the system toward the disease-free state.

In Figure 6.2b, the AIDS-stage population first increases slightly and then decreases over time. The curve for the lower fractional order $q = 0.75$ remains higher for longer, while the ordinary case $q = 1.00$ decreases faster. This means that stronger memory effects can delay the reduction of the AIDS-stage population.

The phase graph in Figure 6.2c shows the relationship between the infected and treated compartments. The trajectories move through a similar region of the phase plane, but the curves separate according to the fractional order. This indicates that changing q

affects not only the size of each compartment but also the dynamic relationship between infection and treatment.

Figure 6.2d shows the fractional memory kernel. The ordinary exponential case decreases faster, while the Mittag–Leffler type curves associated with fractional orders decay more slowly. This supports the interpretation that fractional models preserve past effects for a longer period than classical ordinary differential equation models.

Table 6.2: Comparison of simulation outcomes for different fractional orders.

q	Peak I	Time of peak	Final I	Final T	Final A
1.00	150.00	0.00	2.11	78.78	2.79
0.95	150.00	0.00	3.11	94.27	3.59
0.85	150.00	0.00	6.03	120.99	5.47
0.75	150.00	0.00	10.79	140.94	7.90

The results show that decreasing the fractional order changes the long-term behaviour of the system. For $q = 1.00$, the final infected population is approximately 2.11, while for $q = 0.75$ it increases to approximately 10.79. Similarly, the treated and AIDS-stage populations are higher for smaller values of q. This indicates that stronger memory effects slow the rate at which the system approaches the controlled state. In epidemiological terms, this supports the idea that past behavioural patterns, treatment history, and adherence behaviour may continue to influence HIV dynamics over time.

6.3 Effect of Single Interventions

This section examines the effect of each intervention when applied separately. The purpose is to identify how awareness, safer sexual behaviour, testing and treatment-seeking, and adherence influence the model before they are combined. Four single-intervention scenarios were considered:

awareness only, safer behaviour only, testing only, adherence only.

Insert Figure 6.4 here

Scenario comparison graph showing $I(t)$ under the four single-intervention scenarios.

Figure 6.3: Effect of single interventions on the infected population.

Table 6.3: Single-intervention scenario results.

Scenario	$\mathcal{R}_0$	Peak I	Final I	Final A
Awareness only	0.394	150.00	3.03	5.24
Safer behaviour only	0.394	150.00	3.03	5.24
Testing and treatment-seeking	1.096	150.00	53.60	43.90
Treatment adherence only	1.659	286.44	286.44	99.45

The awareness-only and safer-behaviour-only scenarios both reduce the effective transmission rate and give $\mathcal{R}_0 = 0.394$, which is below the epidemic threshold. This shows that interventions acting directly on transmission can strongly reduce the reproduction number. Testing and treatment-seeking alone improves movement from the infected class to the treated class, but it does not directly reduce transmission in the same way as awareness or safer behaviour. As a result, the reproduction number remains above one in the testing-only case. Adherence support alone reduces progression from treatment to AIDS, but it does not sufficiently reduce new infections when used in isolation; therefore, the model still shows persistence under that scenario.

These results show that single interventions can be useful, but their effectiveness depends on which model pathway they influence. Interventions that reduce transmission have a stronger direct effect on $\mathcal{R}_0$, while treatment-related interventions mainly influence the distribution between infected, treated, and AIDS-stage compartments.

6.4 Combined Intervention Strategy

The combined intervention strategy applies several forms of control at the same time. This is important because HIV transmission is not controlled by one factor only. Awareness, safer behaviour, testing, treatment access, and adherence all contribute to the long-term epidemic outcome.

Three combined scenarios were simulated: a moderate combined intervention, the main combined intervention, and a strong combined intervention. The results are shown in Table 6.4.

Insert Figure 6.5 here

Scenario comparison graph showing $I(t)$ for no intervention, single interventions, and combined intervention strategies.

Figure 6.4: Comparison of single and combined intervention strategies.

Table 6.4: Combined-intervention scenario results.

Scenario	$\mathcal{R}_0$	Peak I	Final I	Final A
Combined intervention	0.137	150.00	1.03	1.80
Combined moderate ($u = 0.5$)	0.332	150.00	2.21	3.95
Strong combined ($u = 0.8$)	0.060	150.00	0.64	1.61

The combined intervention strategies produce the strongest control results. The main combined intervention gives $\mathcal{R}_0 = 0.137$, while the strong combined intervention gives $\mathcal{R}_0 = 0.060$. Both values are far below one, indicating strong epidemic control. The final infected population also decreases substantially, reaching approximately 1.03 under the main combined intervention and 0.64 under the strong combined intervention.

These results support the conclusion that combined interventions are more effective than isolated intervention strategies. In particular, reducing transmission while also improving testing, treatment uptake, and adherence produces better long-term control than focusing on one pathway alone.

6.5 Sensitivity Analysis Results

Sensitivity analysis was used to identify the parameters and intervention controls that have the strongest influence on the basic reproduction number. The normalized sensitivity index measures the proportional change in $\mathcal{R}_0$ caused by a proportional change in a parameter. A positive sensitivity index means that increasing the parameter increases $\mathcal{R}_0$, while a negative index means that increasing the parameter reduces $\mathcal{R}_0$.

Insert Figure 6.6 here

Sensitivity bar chart from the dashboard showing normalized sensitivity indices of $\mathcal{R}_0$.

Figure 6.5: Normalized sensitivity indices for $\mathcal{R}_0$.

Table 6.5: Leading normalized sensitivity indices for $\mathcal{R}_0$.

Parameter or control	Sensitivity index
u_2	-1.000
β_0	1.000
u_1	-0.667
η	0.525
μ	-0.407
u_4	0.380
δ	-0.227
τ	-0.203

The baseline transmission rate β_0 has a positive sensitivity index of approximately 1.000, which means that increasing the baseline transmission rate increases $\mathcal{R}_0$ almost proportionally. The safer-behaviour intervention u_2 has a sensitivity index of approximately -1.000, showing that increasing safer behaviour strongly reduces $\mathcal{R}_0$. Awareness u_1 also has a negative sensitivity index, confirming its role in reducing effective transmission.

The sensitivity results therefore support the intervention findings: reducing transmission is central to controlling the epidemic threshold. Treatment and adherence parameters also affect the system, especially by changing movement into treatment and progression to AIDS, but transmission-related factors are the strongest drivers of $\mathcal{R}_0$ in this configuration.

6.6 Dashboard Demonstration

The computational part of this thesis was implemented as an interactive Flask dashboard. The dashboard allows the user to adjust initial conditions, biological parameters, fractional order, and intervention levels. It then displays time-series graphs, $\mathcal{R}_0$, stability interpretation, scenario comparison, memory-effect comparison, sensitivity analysis, and export tools.

Insert Figure 6.7 here

Screenshot of the dashboard baseline simulation page.

Figure 6.6: Dashboard baseline simulation interface.

Insert Figure 6.8 here

Screenshot of the scenario comparison and memory-effect dashboard tabs.

Figure 6.7: Dashboard scenario and memory-effect visualisation.

Insert Figure 6.9 here

Screenshot of the sensitivity analysis and Chapter 6 result generator tabs.

Figure 6.8: Dashboard sensitivity analysis and thesis-result generator.

The dashboard supports the thesis in three main ways. First, it provides visual evidence for the analytical model by plotting the numerical trajectories of the four compartments. Second, it allows intervention scenarios to be compared using the same baseline parameter values. Third, it provides export tools for CSV tables, figures, and thesis-ready interpretation text. This makes the mathematical model easier to communicate during thesis writing and presentation.

6.7 Public Health Interpretation

The numerical results have direct public-health interpretation, although the model is intended for academic simulation rather than clinical prediction. The results show that interventions reducing effective transmission have a strong effect on the reproduction number. In practical terms, awareness campaigns, risk-reduction education, condom use, and safer sexual behaviour can reduce the probability that susceptible individuals become infected.

Testing and treatment-seeking interventions are also important because they move infected individuals into treatment more quickly. This reduces the untreated infected population and increases the treated population. Treatment adherence reduces progression from treatment to AIDS, helping to keep the AIDS-stage population lower. These findings are consistent with the idea that HIV control requires both prevention and treatment support.

In the context of Rwanda's HIV response, the model supports the importance of integrated intervention programmes. Rwanda has made progress in HIV testing, treatment, and viral suppression, but continued prevention and adherence support remain important. The combined-intervention simulations show that applying behavioural and biomedical interventions together produces stronger control than applying one intervention alone.

The fractional-order results also provide an important interpretation. When $q < 1$, past behaviour and treatment history influence present dynamics through memory effects. This reflects the fact that public health interventions may not act instantaneously; instead, their influence may persist over time through community awareness, adherence habits, stigma reduction, and accumulated treatment experience.

6.8 Chapter Summary

This chapter presented numerical results for the fractional-order SITA HIV model with social behaviour interventions. The baseline simulation produced $\mathcal{R}_0 = 0.430$, indicating a controlled epidemic status under the selected intervention-adjusted parameters. The memory-effect comparison showed that decreasing the fractional order changes the long-term trajectories of the infected, treated, and AIDS-stage populations. Single-intervention simulations showed that transmission-reduction interventions have a strong direct effect on $\mathcal{R}_0$, while combined interventions produced the strongest overall control. Sensitivity analysis confirmed that the baseline transmission rate and transmission-reduction controls are among the most influential quantities. Finally, the dashboard demonstration showed how the mathematical model can be explored interactively and used to generate thesis-ready graphs, tables, and interpretations.

Chapter 7

Conclusion and Recommendations

7.1 Conclusion

This thesis develops a fractional-order SITA HIV transmission model incorporating social behaviour interventions. The use of Caputo fractional derivatives allows the model to capture memory effects, while the Gamma and Mittag-Leffler functions provide the mathematical foundation for the fractional framework. The model connects HIV transmission, treatment initiation, AIDS progression, and behavioural interventions in a single mathematical structure.

The theoretical analysis focuses on positivity, boundedness, disease-free equilibrium, the basic reproduction number, and fractional-order stability. The computational component demonstrates how fractional dynamics and social behaviour interventions can be explored through simulation and dashboard visualisation.

7.2 Recommendations

1. Future work may include age or gender structure.
2. Future studies may estimate parameters using Rwanda-specific data.
3. Stochastic fractional models may be considered for small populations.
4. The dashboard may be extended into a decision-support educational tool.
5. Other fractional operators, such as Caputo–Fabrizio or Atangana–Baleanu derivatives, may be compared with the Caputo derivative.

7.3 Limitations of the Study

1. The model assumes homogeneous mixing.
2. Parameter values may be literature-based or assumed.
3. The model is deterministic and does not include random effects.
4. The intervention functions simplify complex human behaviour into bounded mathematical controls.
5. The dashboard is for academic simulation, not clinical prediction.

Chapter A

Appendix A: Fractional Adams–Bashforth–Moulton Solver Skeleton

```
import numpy as np
from scipy.special import gamma


def fractional_abm_solver(rhs, y0, t, q, params):
    """Predictor-corrector␣solver␣for␣Caputo␣fractional␣systems
        ."""
    n = len(t)
    h = t[1] - t[0]
    y = np.zeros((n, len(y0)), dtype=float)
    y[0] = np.maximum(y0, 0.0)
    fvals = np.zeros_like(y)
    fvals[0] = rhs(t[0], y[0], params)

    for k in range(1, n):
        history = np.arange(k, 0, -1, dtype=float)
        b = history**q - (history - 1.0)**q
        y_pred = y[0] + (h**q / gamma(q + 1.0)) * (b @ fvals[:k
            ])

        a = np.empty(k, dtype=float)
        a[0] = (k - 1)**(q + 1) - (k - 1 - q) * k**q
        if k > 1:
            m = np.arange(k - 1, 0, -1, dtype=float)
            a[1:] = (m + 1.0)**(q + 1) + (m - 1.0)**(q + 1) -
                2.0*m**(q + 1)

        y[k] = y[0] + (h**q / gamma(q + 2.0)) * (
            rhs(t[k], y_pred, params) + (a @ fvals[:k])
```

```
        )
        y[k] = np.maximum(y[k], 0.0)
        fvals[k] = rhs(t[k], y[k], params)

    return y
```

Chapter B

Appendix B: Proposed Baseline Parameter Table

Parameter	Description	Baseline Status
β_0	Baseline transmission rate	Literature/assumed
μ	Natural death rate	Estimated
τ	Treatment initiation rate	Literature/assumed
δ	Progression from infected to AIDS	Literature/assumed
ρ	Progression from treated to AIDS	Literature/assumed
d	AIDS-induced mortality	Literature/assumed
η	Relative infectiousness under treatment	Literature/assumed
q	Fractional order	Simulation parameter
u_1, u_2, u_3, u_4	Social behaviour interventions	User-defined

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